① Aspect 2 (Highest Relevance)

Identify whether a mathematical result or conclusion makes sense given the context

In *memory-stick-qs1*, the core focus is not merely on performing calculations, but on determining whether results such as **storage capacity, number of files, or remaining space** are **realistically reasonable**.

Students must judge whether an answer that is mathematically correct is **practically possible**, taking into account:

- Storage units (KB / MB / GB)
- Actual file sizes
- Physical and practical limits of the storage device

This directly reflects **reasonableness and contextual validity**, making Aspect 2 the most strongly represented.

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② Aspect 1

Interpret a mathematical result back into the real-world context

The questions require students to:

- Convert a numerical result (e.g. "23.6 files")
- Into a meaningful real-world conclusion (e.g. "only 23 files can be stored")

This highlights that:

- A mathematical result is not automatically the final answer
- The result must be interpreted within the **memory stick usage context**

③ Aspect 9

Understand the relationship between context and mathematical representation

The task involves linking:

- Abstract representations (numbers, calculations)
- With the real-world situation (files and storage space)

Students must understand:

- Why rounding down is required
- Why partial files cannot be stored
- How mathematical representations are constrained by context

④ Aspect 3

Identify the limits of the model used

The problem relies on implicit simplifying assumptions, such as:

- All files being the same size
- No system or formatting space being used
- No variation in file types

Higher-level responses may recognise that:

- This is a **simplified mathematical model**
- Actual usable storage is usually less than the advertised capacity

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⑤ Aspect 6

Use real-world knowledge to adjust or contextualise results

More advanced interpretations may incorporate:

- Formatting losses
- File system overhead
- Differences between nominal and usable storage capacity

This demonstrates the application of **real-world considerations to refine mathematical conclusions**.

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⑥ Aspect 7

Construct and communicate explanations and arguments in context

The questions often require students to:

- Explain *why* a conclusion is valid
- Present reasoning in clear, complete sentences

This emphasises that students must:

- Do more than compute an answer
- Communicate reasoning in a way that is meaningful beyond pure mathematics

① Aspect 2（相关性最高）

Identify whether a mathematical result … makes sense given the context

📌 在 memory-stick-qs1 中，核心问题并不是“会不会算”，而是：

- 得到的**存储容量 / 文件数量 / 剩余空间**是否**符合现实**
- 是否出现“数学上对、现实中不可能”的情况

👉 学生必须结合：

- 存储单位（KB / MB / GB）
- 实际文件大小
- 存储设备的物理限制

来判断答案是否**合理**，这是典型的 **合理性判断（reasonableness）**。

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② Aspect 1

Interpret a mathematical result back into the real world context

📌 题目要求学生：

- 把计算结果（如“可存 23.6 个文件”）
- **重新解释为现实结论**（如“最多只能存 23 个文件”）

👉 这里强调：

- 数学结果 ≠ 最终答案
- 必须**回到 memory stick 的使用情境中解释**

③ Aspect 9

Understand the relationship between context and representation

📌 memory-stick-qs1 涉及：

- 抽象表示（数值、算式）
- 与具体情境（文件、存储空间）的对应关系

👉 学生需要理解：

- 为什么要向下取整
- 为什么不能“存一半文件”
- 数学表示如何受现实情境约束

④ Aspect 3

Identify the limits of the model used

📌 题目隐含的模型假设包括：

- 所有文件大小一致
- 不考虑系统占用空间
- 不考虑文件格式差异

👉 高水平回答会意识到：

- 这是一个**简化模型**
- 实际 memory stick 的容量并非完全可用

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⑤ Aspect 6

Use knowledge of real-world impacts to adjust results

📌 在更深入的解释中，学生可能提到：

- 格式化损耗
- 文件系统占用
- 实际可用容量小于标称容量

👉 这体现了用**现实知识修正或限定数学结论。**

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⑥ Aspect 7

Construct and communicate explanations in context

📌 题目通常要求：

- 说明“为什么”
- 用完整语句解释结论

👉 不只是算出答案，而是：

- 清楚说明推理过程
- 让结果对“非数学背景的人”也有意义